Bachelor of Information Technology in Computer Systems Security (BCSS)

IPB49804 Final Year Project 1

Final Year Project Proposal Form

Final Year Project's Proposal Form (fill up by student)		
1.	Student's Name	QASEM FARES HESHAM ABDO
2.	Contact Number	0143085661
3.	Current CGPA	2.69
4.	Working Title	Personal Data Exposure Checker
5.	Background/Introduction	With the growing number of online accounts and cloud-based services, users face increasing cybersecurity risks such as identity theft, phishing, and unauthorized data disclosure. Many individuals are unaware that their personal information (such as emails or usernames) may already be exposed through data breaches. This project aims to develop a *Personal Data Exposure Checker*, a web-based application that allows users to check whether their personal identifiers appear in publicly known data breaches. The system will focus on user privacy, transparency, and cybersecurity awareness by using open and verified breach data sources instead of commercial APIs.
6.	Problem Statement	Many individuals remain unaware that their personal data may have been leaked in past data breaches. Existing exposure checkers often depend on proprietary APIs or require users to submit sensitive information without transparency about data handling. There is a need for a transparent, privacy-focused solution that allows users to check for data exposure securely using open, ethical, and research-friendly datasets. This project addresses that gap by developing a safe and educational tool for users to assess their cyber risk without compromising privacy.
7.	Objectives	1. To study and analyze existing data exposure and breach detection techniques. 2. To develop a web application that checks user identifiers (e.g., emails, usernames) against public or open-source breach datasets through secure API integration. 3. To evaluate the system's functionality, usability, and privacy protection effectiveness.

8.	Project Scope / Coverage	1. Target Users: General internet users who wish to verify if their data has appeared in known breaches. 2. System Functions: - Input email, username, or phone number for exposure checking. - Query open or research-friendly APIs (e.g., DeHashed, LeakCheck, Scylla.so) or local datasets. - Display results with breach details and cybersecurity advice. - Ensure user privacy by hashing and not storing sensitive inputs. 3. Platform: Web-based system using React.js (frontend) and Node.js/Express (backend).
9.	Literature Review (Proposal only - 3 articles)	1. Prior research shows that breach lookup services (e.g., DeHashed, Scylla.so) can safely provide exposure data without storing user information. 2. Techniques such as client-side hashing, anonymization, and encrypted API communication enhance privacy in breach detection systems. 3. Studies highlight the importance of transparency and ethical data handling in cybersecurity applications.
10.	Contribution to Cybersecurity Domain	- Enhances cybersecurity awareness by allowing users to safely check their data exposure status. - Promotes privacy-preserving system design in cybersecurity applications. - Demonstrates integration of secure web development and ethical data handling practices.
11.	Proposed Supervisor Signature	

For Office Use Only

1.	Evaluator's Comment:	☐ Accept ☐ Reject
2.	Comments/ Remarks	
3.	Assigned supervisor	
4.	Checked & Verified by: ..... Name: Designation: Date:	